

Principles and Approach to Developing Mammalian Cell Culture Media for High Cell Density Perfusion Process Leveraging Established Fed-Batch Media

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Perfusion medium was successfully developed based on our fed-batch platform basal and feed media. A systematic development approach was undertaken by first optimizing the ratios of fed-batch basal and feed media followed by targeted removal of unnecessary and redundant components. With this reduction in components, the medium could then be further concentrated by 2× to increase medium depth. The medium osmolality was also optimized where we found ~360 mOsm/kg was desirable resulting in a residual culture osmolality of ~300 mOsm/kg for our cell lines. Further building on this, the amino acids Q, E, N, and D were rebalanced to reduce lactate and ammonium levels, and increase the cell-specific productivity without compromising on cell viability while leaving viable cell density largely unaffected. Further modifications were also made by increasing certain important vitamin and lipid concentrations, while eliminating other unnecessary vitamins. Overall, an effective perfusion medium was developed with all components remaining in the formulation understood to be important and their concentrations increased to improve medium depth. The critical cell-specific perfusion rate using this medium was then established for a cell line of interest to be 0.075 nL/cell-day yielding 1.2 g/L-day at steady state. This perfusion process was then successfully scaled up to a 100 L single-use bioreactor with an ATF6 demonstrating similar performance as a 2 L bioreactor with an ATF2. Large volume handling challenges in our fed-batch facility were overcome by developing a liquid medium version of the powder medium product contained in custom totes for plug-and-play use with the bioreactor.

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Introduction

Continuous processing for biopharmaceutical manufacturing has gained significant attention in the last several years as a means of process intensification to significantly improve monoclonal antibody (mAb) production over the conventional fed-batch process. Perfusion process is an integral part of continuous processing. Volumetric productivity of perfusion cultures in the range of 1–2 g/L-day is now commonplace; for example, Xu and Chen, 2016.¹ Moreover, 3–5 g/L-day range has now been recently reported by several companies at the ECI Integrated Continuous Biomanufacturing

Conference (2015) and Cell Culture Engineering Conference (2016). As an example, an intensified perfusion process running at 2 g/L-day for 10 harvest days would already achieve a fed-batch titer equivalent of 20 g/L as compared to typical fed-batch titers in the 3–5 g/L range. However, it is more likely that this process would run for longer to maximize plant production time, so in such a case, titer would far exceed beyond the capability of any known fed-batch process. In the future, when 3–5 g/L-day will become routinely obtainable, such productivity would drastically reduce the commercial scale bioreactor footprint of today. Walther et al. showed using costs of goods comparison based on net present value analysis that using a completely integrated continuous process both upstream and downstream can reduce both capital and operating cost by an average of 55% when compared to a conventional fed-batch process.² Additionally, Klutz et al. presented a concept of “biofacility of

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the future” where continuous processing is an indispensable methodology to make the facility smaller, faster, and less expensive due to the higher yields of perfusion cultures.³ Feasibility of fully integrated upstream and downstream continuous processing has also been successfully demonstrated by Warikoo et al. and Godawat et al.^{4,5} The complexity of perfusion (e.g., cell retention device, long run duration, and higher contamination risk) historically has been justified as necessary for the production of complex, labile products such as recombinant blood clotting Factor VIII and enzyme therapies.^{6–9} This is not to say mAbs have not been produced in the past utilizing perfusion process. Notably, Janssen Biotech has several mAb products on the market that are produced using perfusion technology.⁹ The history of the adoption of the perfusion process may possibly be due to the need to overcome low titers and also institutional know-how at the time of their mAb development. Contrary to what has been described above, Janssen Biotech has since reverted to fed-batch processes for its later products, possibly due to simpler operation and sufficient titer. Despite such a case, companies such as Amgen, Merck and Pfizer with long-established fed-batch processes (source from 2015 ECI Conference on Integrated Continuous Biomanufacturing in Berkeley, CA) are exploring perfusion as a disruptive method to substantially increase the yield of mAbs beyond the realm of the fed-batch capability.

For traditional fed-batch companies such as Boehringer Ingelheim, developing a perfusion process without prior technology requires leveraging the existing fed-batch platform knowledge. Taking such a route to evolve a medium for perfusion relies on designing optimal mixtures of basal and feed media that can serve as a strong baseline for starting perfusion medium development. One area of important focus for perfusion medium is to control and minimize unnecessary cell growth. Doing so could reduce the cell bleed rate commonly used for controlling cell density and, hence, minimize the product loss from cell bleed. Cell bleed rate can typically constitute 10–30% of the perfusion rate, so minimizing cell bleed would improve the overall yield. Another potential benefit of suppressing cell growth in perfusion culture is improvement of cell-specific productivity (q_p). High cell density can typically be achieved with mere perfusion cell retention device. Preventing runaway cell growth can help channel more carbon and energy source toward secondary metabolism and biosynthesis of the product. Work by Du et al. showed that by adding a small molecule inhibitor of cyclin-dependent kinases (CDK) 4/6 to the medium, Chinese Hamster Ovary (CHO) cells could be arrested in the G0/G1 phase of the cell cycle.¹⁰ This in turn led to significant reduction and control of cell proliferation with a concomitant increase in specific productivity. In one cell line, q_p increased $2\times$ to as high as ~ 110 pg/cell-day. Another aspect for perfusion medium is to enhance the medium depth. This can be achieved by removing redundant and unnecessary components and further increasing the concentration of the medium. The greater the medium depth, the lower the perfusion rate needed to maintain a high viability steady-state culture resulting in lower medium utilization. Other challenges surrounding perfusion’s large volume medium requirement are facility preparation equipment capacity, storage, and liquid handling and transfer capability. These are especially true for a facility designed for only fed-batch processes. Large medium preparation vessels are not available to pair with many folds smaller bioreactors, as this is usually the

magnitude difference in ratio between medium vessels and production bioreactors for perfusion. Sufficient cold storage capacity for large volume perfusion medium is also not typically available in a fed-batch facility. There is also insufficient floor space to shuttle around large volumes. Creative methods need to be undertaken to manage large bulk medium volume to make perfusion processes fit for existing fed-batch facilities. Development of a liquid medium product direct from the vendor and ready to use was a strategy that we employed to address the above concerns.

The work in this article shares the examples of approaching medium development for perfusion starting with fed-batch media. Key development parameters such as medium osmolality, nutrient components, and cell-specific perfusion rate are discussed. Proof-of-concept of medium improvement is demonstrated in 2 L bioreactor using the alternating tangential flow (ATF) system as cell retention device. Scale-up and the approach for implementation into our fed-batch pilot plant facility in the 100 L single-use bioreactor are also addressed.

Materials and Methods

Cell line and media

In-house proprietary recombinant Chinese Hamster Ovary (CHO) cell lines producing monoclonal antibodies were used. Thawed inoculum cultures were grown in a proprietary chemically defined medium and maintained in shake flasks until the target volume and cell density were reached. At such time, cells were committed to either bioreactors or deep-well plate experimentation. The media used in the perfusion phase of cell culture consisted of a combination of the in-house basal and feed media previously established as part of Boehringer Ingelheim’s fed-batch platform with the modifications discussed below. Cells for deep-well plate work were inoculated directly into these perfusion media, whereas bioreactor cultures were inoculated in a diluted ($\sim 85\%$) version of the media to adjust osmolality for the differences in inoculation cell density.

Deep-well plate model of perfusion

Oxygen 24-well presterilized deep-well plates (Neta Scientific Inc, Hainesport, NJ) were used. A starting cell density of 20×10^6 cells/mL was targeted for deep-well plate model work to simulate the high cell density growth phase of perfusion and test medium depth. To achieve such high cell densities from shake flask culture, the necessary volume was centrifuged at 1000 rpm for 5 min and the cell pellet resuspended in perfusion media to the desired density before distribution into deep-well plates. The working volume per well was 3 mL. Cells were grown at 33°C with 5% CO₂, 80% humidity, and 200 rpm in an incubator with a 5.0 cm orbit diameter. Medium exchanges were performed daily by centrifuging the plate at 1200 rpm for 5 min, removing supernatant, and resuspending cells in fresh media. Exchange rates between 1/3 and 1 volume medium per volume bioreactor per day (vvd) were tested. A 0.1 mL sample containing cells was taken for cell count daily prior to the exchange. Spent media analysis was performed on the supernatant samples. Similar deep-well plate methods to emulate perfusion cell density cultures were also described by Huang et al. and Du et al.^{10,11} Such a model is intended for highthroughput media screening under a high cell density environment,

where it is normally not obtainable with a standard batch culture. The model does not represent the steady-state nature of a perfusion process, but only the cell growth phase of the perfusion. It has been shown in this work and previous published work to be an effective tool for perfusion medium development.

Bioreactor

The hollow fiber used in all perfusion bioreactors was made of polyethersulfone (PES) with 0.2 μm pore size and 1 mm lumen ID. The recirculation device selected was an ATF2 system for all bench scale work and an ATF6 for the 100 L scale. Both devices were purchased from Repligen (Waltham, MA). All benchtop studies were conducted in 3 L glass stirred tank bioreactors at a 2 L working volume (Applikon Biotechnology, Delft, Netherlands). A customized single-use bioreactor modified for perfusion cell culture (Thermo Scientific, Logan, UT) was used for the 100 L work. Monitoring and control of all bioreactor parameters were performed using a Finesse (Finesse Solutions, Inc., Santa Clara, CA) G3Lab Universal controller. Targeted shear through the hollow fiber was kept below 2000 s^{-1} and matched across all scales.

Process analytics

Viability and cell density were measured by Vi-CELL Cell Counter (Beckman Coulter Life Sciences, Indianapolis, IN). Titer, metabolites, and lactate dehydrogenase (LDH) activity were measured using a Cedex BioHT (Roche Diagnostics GmbH, Mannheim, Germany). These titer measurements had been previously verified to be in agreement with in-house HPLC Protein A methods. Osmolality was measured with a 2020 Multi-Sample Micro Osmometer (Advanced Instruments, Inc, Norwood, MA) based on freezing point depression. Amino acid analysis was performed using an ACQUITY UPLC system (Waters, Milford, MA).

Results and Discussion

Establishment of perfusion medium based on fed-batch basal and feed media

Established fed-batch basal and feed media can be leveraged to develop new perfusion medium rather than starting from first principles. Perfusion medium can be created based on mixing different ratios of fed-batch basal to feed media. This is an efficient and effective strategy since the compositions of the fed-batch media is already demonstrated to support production in cell culture. This approach can be based on existing fed-batch platform media or applying mixtures of different vendor basal and feed media. One important consideration when testing different mixtures of basal and feed media is matching the level of viable cell density (VCD) expected in perfusion with the right osmolality strength of the medium. One can expect at least more than 30×10^6 cells/mL in highly intensified perfusion cultures, but typically $50\text{--}100 \times 10^6$ cells/mL is possible.^{1,4} In some cases, significantly $>100 \times 10^6$ cells/mL have been achieved.^{12,13} Having the right osmolality level to pair with VCD level is important. Osmolality is an indirect indicator for nutrient strength. Having too low a medium osmolality will result in the need for much higher perfusion rates to prevent culture starvation. Having too high an osmolality can result in

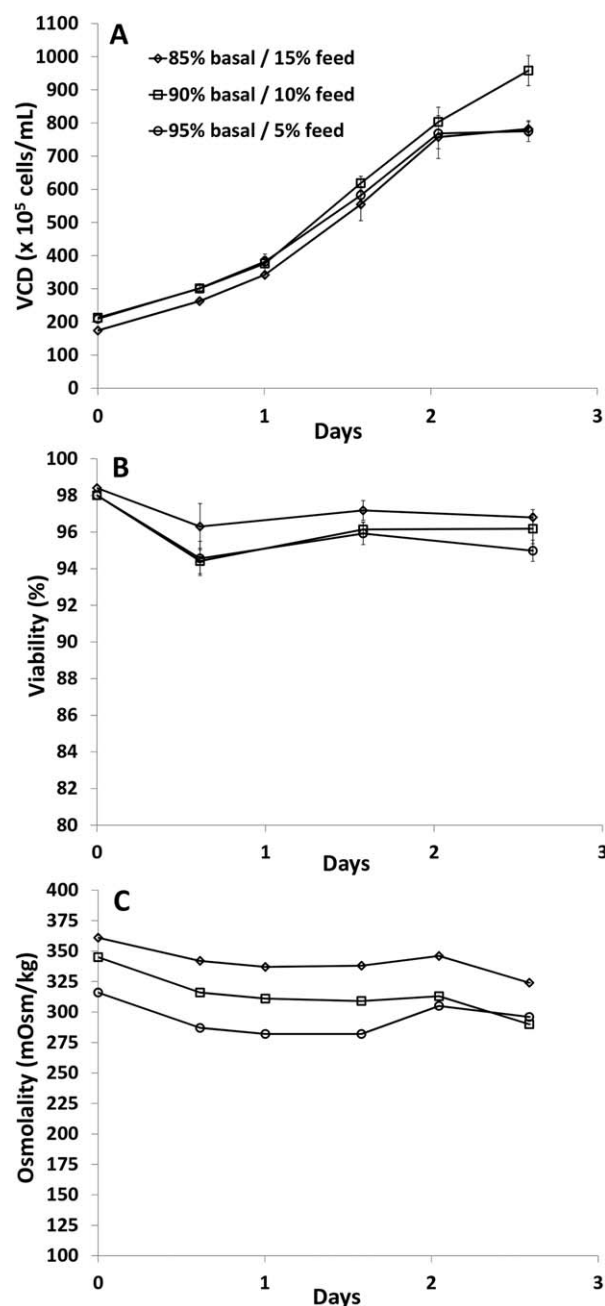


Figure 1. Comparison of different basal-to-feed medium ratios. (A) VCD, (B) cell viability, and (C) culture osmolality.

osmotic stress caused by a higher residual osmolality in the medium and markedly lower nutrient consumption rates due to suppressed cell-specific growth rate. Perfusion rate would need to be adjusted down to balance nutrient consumption and avoid osmolality buildup. Examples of previous work have consistently shown high osmolality decreases cell-specific growth rate, although it can be leveraged to increase the cell-specific productivity.^{14–16} A desirable physiological residual osmolality level for a perfusion culture is ~ 300 mOsm/kg similar to the range of 260–320 mOsm/kg for conventional batch cultures.^{14,17}

Different ratios of basal to feed medium were studied. The ratios included 85:15, 90:10, and 95:5 of basal to feed medium. Figure 1A shows that VCD reached the highest with the 90:10 ratio. Specific growth rate in the exponential phase was also the highest for the 90:10 ratio (0.025 h^{-1})

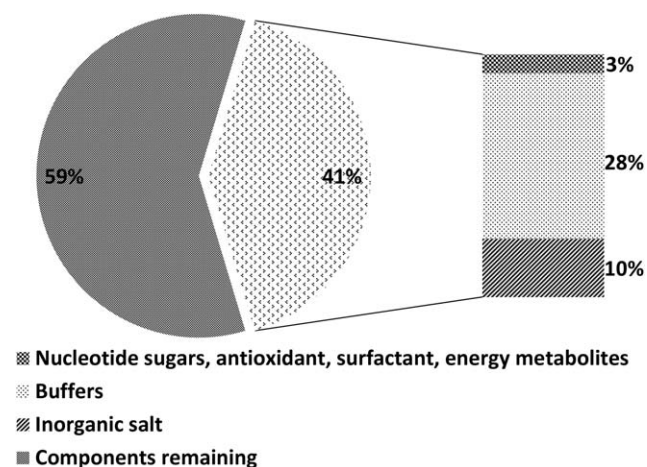


Figure 2. Distribution of the mass of the components in the 90:10 perfusion medium. The 41% is the mass removed and the 59% is the mass remaining. The distribution to the right shows the further breakdown of the classes of components in the 41% removed.

compared to 85:15 (0.022 h^{-1}) and 95:5 (0.019 h^{-1}). No difference in viability was observed within the growth conditions studied (Figure 1B). The residual osmolality was the most favorable for 90:10 being closest to 300 mOsmo/kg (Figure 1C). The 85:15 ratio residual osmolality was too high likely to be inhibitory and the 95:5 ratio was too low for it to be able to support a high-cell-density perfusion culture. Ultimately, the 90:10 ratio with a medium osmolality of $\sim 360 \text{ mOsmo/kg}$ was determined to be the best for supporting high cell density perfusion culture, while capable of maintaining high cell viability. Although the results are based on the deep-well plate model, the relative ranking and our prior experience with this application support this conclusion. From this screening of the basal and feed mixtures, the 90:10 ratio was established as our baseline and first version perfusion medium. From this baseline, further development and optimization of the perfusion medium would be made.

Creation of more chemical space to increase perfusion medium concentration

After establishing the 90:10 ratio fed-batch media mixture as the first version perfusion medium, several components in the mixture were identified as redundant for perfusion culture. For the purpose of increasing nutrient concentration without subjecting the medium to further osmolality increase, 9 components constituting approximately 41% mass of the perfusion medium powder were removed (Figure 2). The powder itself has 75 components. This percentage does not take into account the other single components added to make the final liquid medium. The notable components removed are the buffers HEPES and MOPS, and the salt NaCl which by themselves make up 38% of the powder mass. HEPES and MOPS are deemed unnecessary for buffering capacity of the perfusion medium, and that proper phosphate buffer and bicarbonate concentrations in the medium are sufficient for pH buffering. HEPES has also been reported to be toxic, specifically under light-exposed conditions in the presence of riboflavin.^{17,18} NaCl was occupying undesired osmolality space. Without NaCl, osmolality adjustment can be made with the medium powder itself. The phosphate buffer capacity would also be increased as a result of

removing the components described above and further concentrating the remaining components. Bicarbonate as a separate liquid medium component addition was not changed. With the 9 components removed, 66 components remained and another new version of the perfusion medium powder was created. Several versions of the perfusion medium were made using this powder to evaluate performance. The mass of the powder, hence the remaining components, were increased by 69% to match the original mass concentration; 97% to match the original osmolality; and 125% to reach the solubility limit of the new powder. The other single-component concentrations added to make the liquid medium were left unchanged. These three perfusion media with increased nutrient concentrations were compared to the first version 90:10 ratio perfusion medium.

The results show that with the 69% and 97% increased concentrations, both titer and cell-specific productivity (q_p) were improved compared to the first version 90:10 ratio perfusion medium (Figures 3C,D). The 125% increase showed the most significant improvement in q_p . However, because the cell growth was lower (Figure 3A) due to the higher residual osmolality ($\sim 360 \text{ mOsmo/kg}$) than the other conditions (Figure 3B), the titer was not higher than the 69% and 97% increase in concentration. The 125% increase, however, can be potentially advantageous for suppressing undesired cell growth to limit the use of cell bleed, hence reduce product loss when a sustainable perfusion production system is desired. For the 69% and 97% conditions, the VCD levels were similar to 90:10 medium (Figure 3A); therefore, volumetric productivities were truly higher due to their higher q_p values. The osmolality profiles show that the 97% medium reached closer to the desired physiological osmolality level of 300 mOsmo/kg than the other conditions. The 69% medium had a substantially lower osmolality of 270 mOsmo/kg compared to the 97% and 125% media. This indicates that the 69% medium is too lean and, therefore, may incur higher perfusion rate requirement when used in a perfusion bioreactor. The viability profiles for all four conditions were similar and maintained above 95% (data not shown). Viability was not affected by the concentration differences within the experimental duration tested. Comparing the three concentrations (69%, 97%, and 125%), the 97% increase was selected as the best concentration to adopt for the next version perfusion medium.

Rebalance of energy metabolism amino acids to improve q_p and reduce byproduct waste

The 97% increased concentration perfusion medium was used as the new foundation for targeted medium development of individual components. Rebalancing the amino acid concentrations carried over from the fed-batch basal and feed media mixture was evaluated. Glutamine (Q), glutamic acid (E), asparagine (N), and aspartic acid (D) are important energy metabolism amino acids that play crucial roles not only in biosynthesis but also in the primary metabolism consisting of glycolysis and tricarboxylic acid (TCA) cycle. Figure 4 highlights the various pathway interactions between Q, E, N, and D; some are reversible and others are unidirectional. Modulating the concentrations of these four amino acids can perturb the network in terms of their consumption and production. For fed-batch culture, Q and N are traditionally optimized to support cell growth to increase volumetric productivity. For the perfusion culture, our interest lies more in

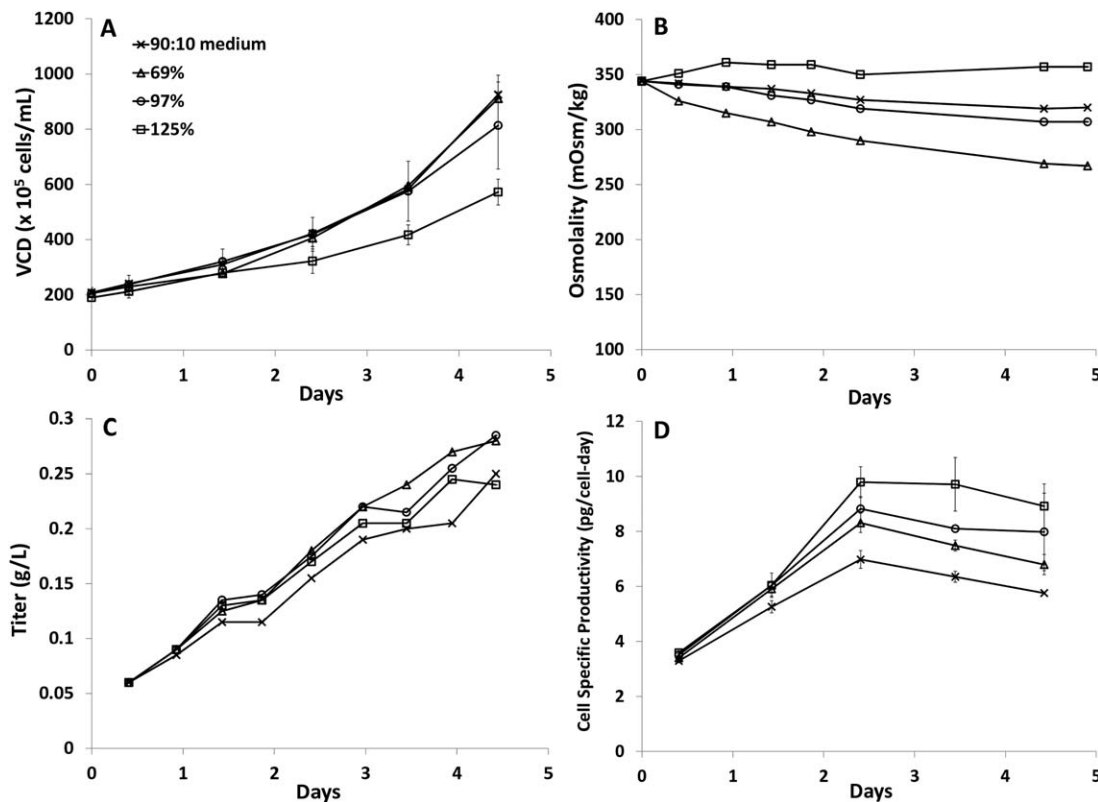


Figure 3. Comparison of higher concentration perfusion media versus the first version 90:10 ratio perfusion medium. The percentages are the mass increase of the reduced component powder in the medium. Titer values shown are measured values before each medium exchange. Cell-specific productivity is calculated based on accumulated titer assuming no product removal from the medium exchange. (A) VCD, (B) culture osmolality, (C) titer, (D) cell-specific productivity.

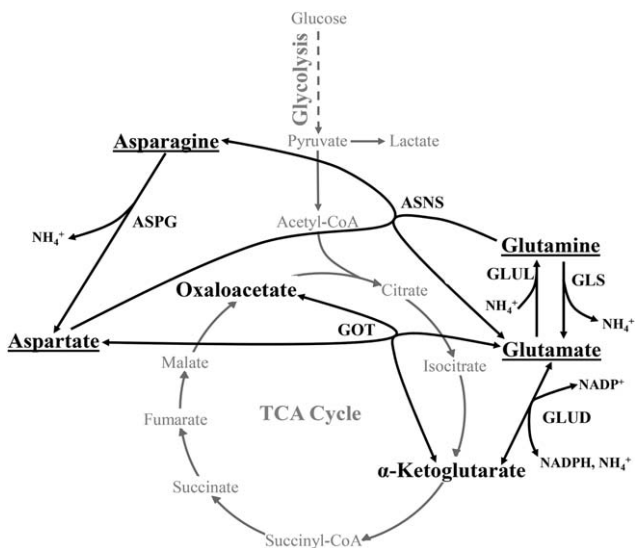


Figure 4. Overview of the glutamine, glutamate, asparagine, and aspartate metabolic pathway interactions. Enzymes involved in these pathways are as follows: asparaginase (ASPG), asparagine synthetase (ASNS), aspartate transaminase (GOT), glutamate dehydrogenase (GLUD), glutamine synthetase (GLUL), and glutaminase (GLS). ASNS reaction is unidirectional converting aspartate and glutamine to asparagine and glutamate. GOT reaction is reversible converting aspartate and α-ketoglutarate to and from glutamate and oxaloacetate. For simplicity, ATP and H₂O are omitted in the diagram of these pathways.

controlling cell growth to shift more energy and carbon sources toward product synthesis. This is the case for perfusion because cell retention device by itself can be leveraged to mechanically concentrate cells to very high cell densities. Under such high cell densities, it would be more important to coerce these cells into diverting energy toward generating product rather than cell mass. In turn, this can enhance q_p , maintain modest cell density without the significant use of cell bleed, and reduce byproduct waste such as lactate and ammonium in the perfusion culture. To perform the rebalance, the total Q, E, N, and D molar concentration is kept constant; however, Q and N concentrations are substantially reduced and the difference in molar equivalent concentrations are made up by adding more E and D, respectively. Even for perfusion media, these amino acids can be sufficient in the single digit millimolar concentration ranges, which are typical of fed-batch basal media. The perfusion rate needed will also depend on the concentrations used. Q was reduced by 89% in the 97% increased concentration medium (the control medium for this study), a level that does not promote cell growth, but instead induces more Q production from E consumption. Rajendra et al. showed that higher product yields could be obtained when using low glutamine concentrations of 0–2 mM for transient HEK-293K and CHO cells.¹⁹ Higher glutamine culture concentration had also been reported to decrease other amino acid consumption.²⁰ This can potentially cause other amino acids to be less efficiently transported and metabolized, hence why lower glutamine in the culture can induce higher consumption of glutamate, and switch to generation rather consumption of

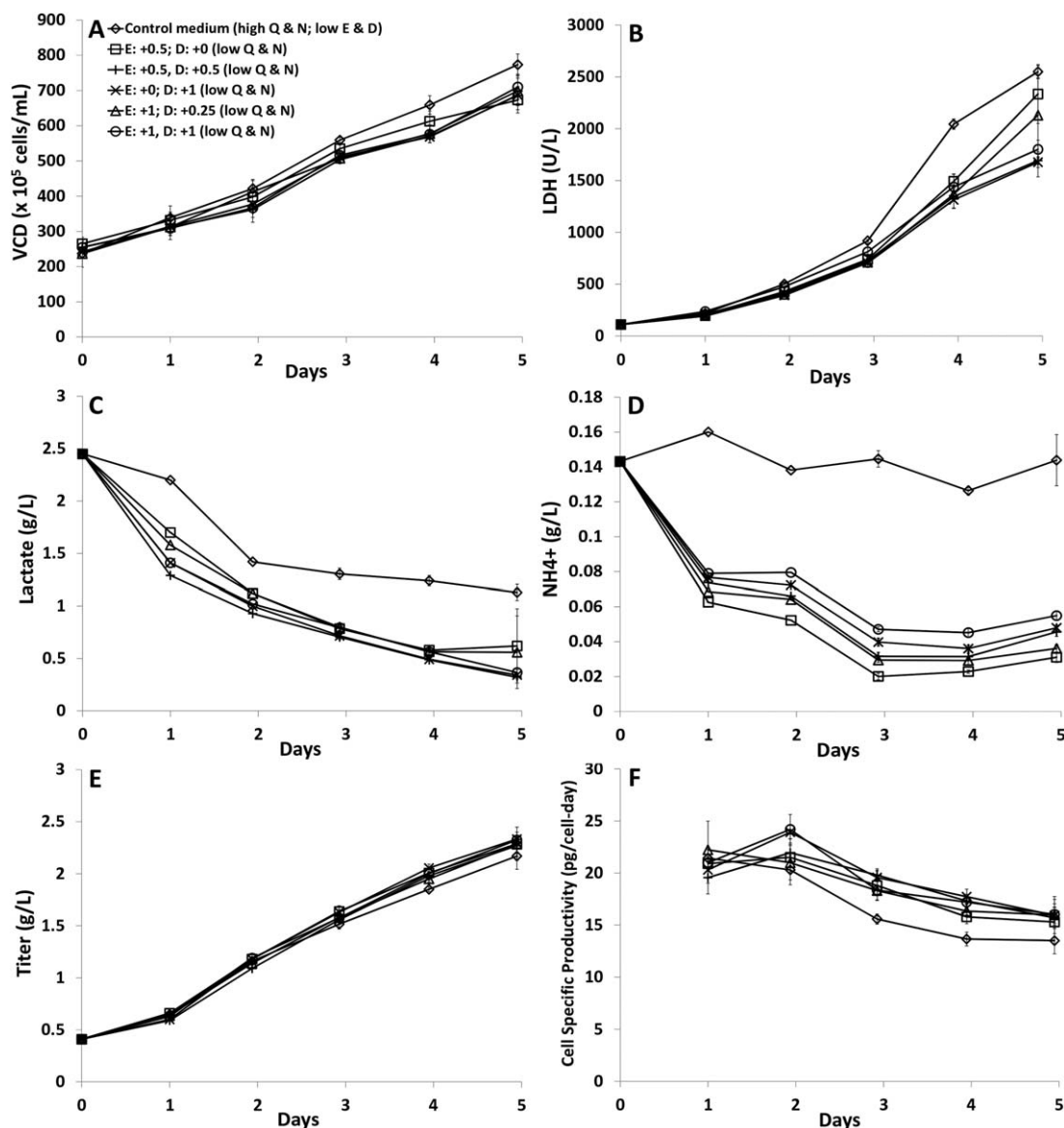


Figure 5. Comparison of different combinations of high and low *E* and *D* concentrations in a low *Q* and *N* background versus the 97% increased concentration control medium. The control medium has a high *Q* and *N*, and low *E* and *D* concentration background. The numbers +1, +0.5, +0.25, and +0 represent the full, half, quarter, and no increase in *E* and *D* concentrations. A full increase for *E* is 81% and for *D* is 83% from the levels in the control medium. (A) VCD, (B) lactate dehydrogenase (LDH) activity in the supernatant, (C) lactate, (D) ammonium, (E) titer, and (F) cell-specific productivity.

glutamine. Another area for further understanding this mechanism may be explained by the mTOR activation and inhibition caused by glutamine influx and efflux, respectively. Active mTOR regulates protein translation and cell growth, and its inhibition leads to activation of autophagy for alternative nitrogen source. Glutamine transport occurs reciprocally with other essential amino acids through the bidirectional antiporter SLC7A5-SLC3A2.²¹ Based on this mechanism, it is possible that drastically altering the extracellular amino acid concentrations will affect the amino acid metabolism through the mTOR pathway. *N* was initially reduced by 67% to prevent futile utilization where no performance benefits were observed, and concentration was also not at a cell growth limiting level. Follstad et al. showed that low *N* concentration below 5 mM could be leveraged to limit cell growth in perfusion cultures.²² However, due to concern of *N* starvation potentially generating *N* to *S* (serine)

sequence variants in the product,^{23,24} the final concentration of *N* was only reduced by 34% in our case. *E* and *D* were increased by 81% and 83%, respectively, to compensate for the molar concentration reduction in *Q* and *N*. The experiments conducted studied combinations of full *E* and *D* increase equivalent to 81% and 83% as described above, half increase, quarter increase, and no increase in the same reduced *Q* and *N* concentration background.

The study was originally designed using design of experiment (DOE) methodology. However, after observing the results, there were no significant differences among any of the conditions receiving additional *E* and/or *D* (Figures 5A–F). The only significant difference occurs when comparing the group to the control medium (97% increased concentration). Under any of the low *Q* and *N* conditions with *E* or *D* added, VCD levels were slightly lower than the control medium (Figure 5A). The lower VCD profiles are expected

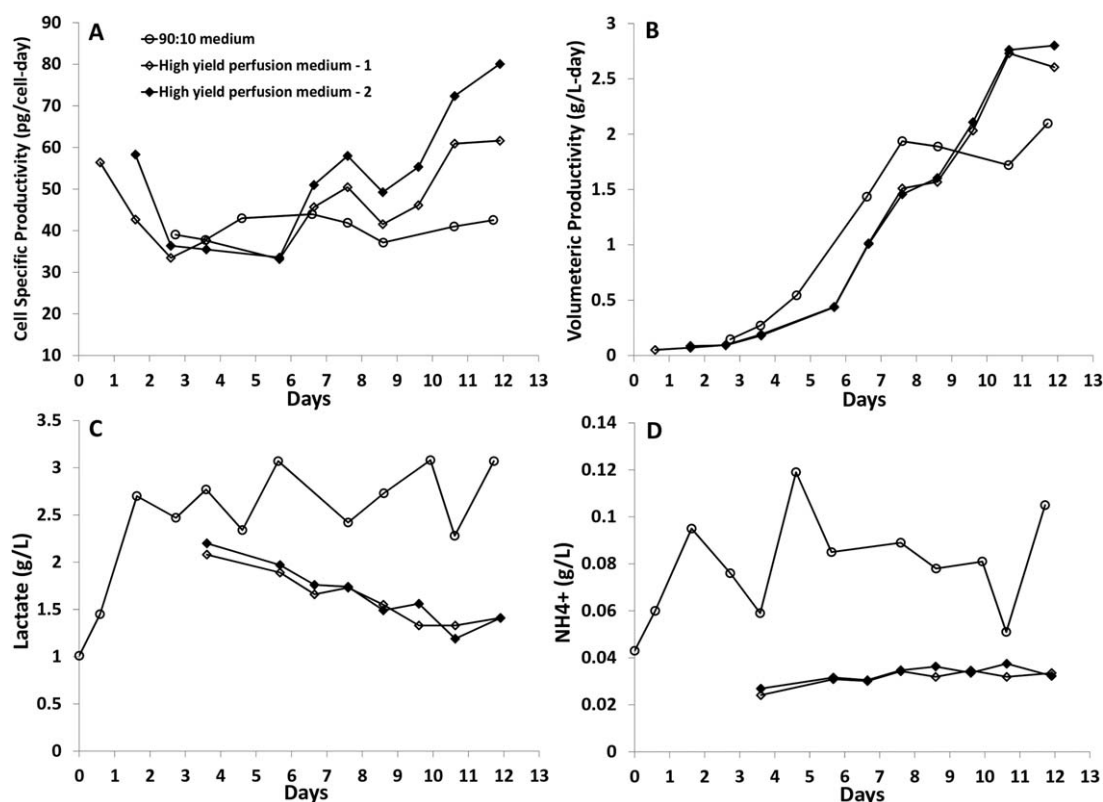


Figure 6. Two-liter ATF bioreactor comparison of the high-yield perfusion medium to the first version 90:10 ratio medium using fed-batch basal and feed. The high-yield perfusion medium was conducted in duplicate runs: (A) cell-specific productivity, (B) volumetric productivity, (C) lactate, and (D) ammonium.

and desired. Viability profiles were very similar among all conditions, all >90% (data not shown). Despite the viability being similar and high, Figure 5B shows that the LDH level in the supernatant is higher for the control medium than the other E and/or D added media. This implies that the cells may actually be less healthy with more compromised membrane integrity in the control medium than the low Q and N, higher E and D media. The extracellular LDH release used in this regard is a well-established biomarker of cell membrane integrity commonly used for quantifying cell death.^{25,26} Furthermore, lactate and ammonium were substantially reduced in all the low Q and N, higher E and D media (Figures 5C,D). Cell-specific productivities were also improved, and the titer levels were also slightly higher than the control medium despite reduced VCD levels (Figures 5E,F). The lower ammonium is attributable to the lower source of glutamine and asparagine where free ammonium is generated from spontaneous glutamine degradation and also from their intracellular enzymatic conversions. The lower lactate is hypothesized to be because of the reduced overflow metabolism where glucose is now the only significant carbon source for glycolysis and TCA cycle. For this reason, cells are coerced to more efficiently utilize the abundant E and D for biosynthesis. Similarly, in other work reported, Altamirano et al. showed glutamine replacement with glutamate in the medium reduces lactate and ammonium in CHO cell cultures.^{27,28}

Following the Q, E, N, and D study, the new levels for E and D were chosen to be the full concentration increases of 81% and 83%, and the Q and N levels were kept at the reduced levels described earlier. Although there were no

significant differences among the E and D concentrations studied, we chose the highest E and D concentrations to ensure levels are sufficient for other cell lines and for use in high-intensity perfusion bioreactors. Other refinements and optimization to the perfusion medium were also made, but results are not discussed in detail in this article. The notable areas include optimizing the L-tyrosine and L-cysteine/L-cystine levels and staying within their solubility limits, and doubling lipid component concentrations such as choline, inositol, linoleic acid, and lipoic acid. Vitamins were also optimized doubling the concentrations of some, while removing other nonessential ones. Overall, removing unnecessary components and increasing the concentrations of important amino acids, lipids, and vitamins improve the medium depth of the perfusion medium and potentially allow lower perfusion rates to be used. The removal of the described components did not show any negative impact on cell culture and increasing the said components showed further improvements in both the volumetric and cell-specific productivities. With all these modifications combined, a high-yield perfusion medium was created.

Demonstrating enhanced performance of high-yield perfusion medium in 2 L ATF bioreactor

The high-yield perfusion medium was evaluated against the first version 90:10 ratio perfusion medium in 2 L bioreactor utilizing ATF2 for perfusion. A different cell line of interest was evaluated instead of the one used for medium development in the deep well plate. Same perfusion conditions were used for both media. The high yield perfusion

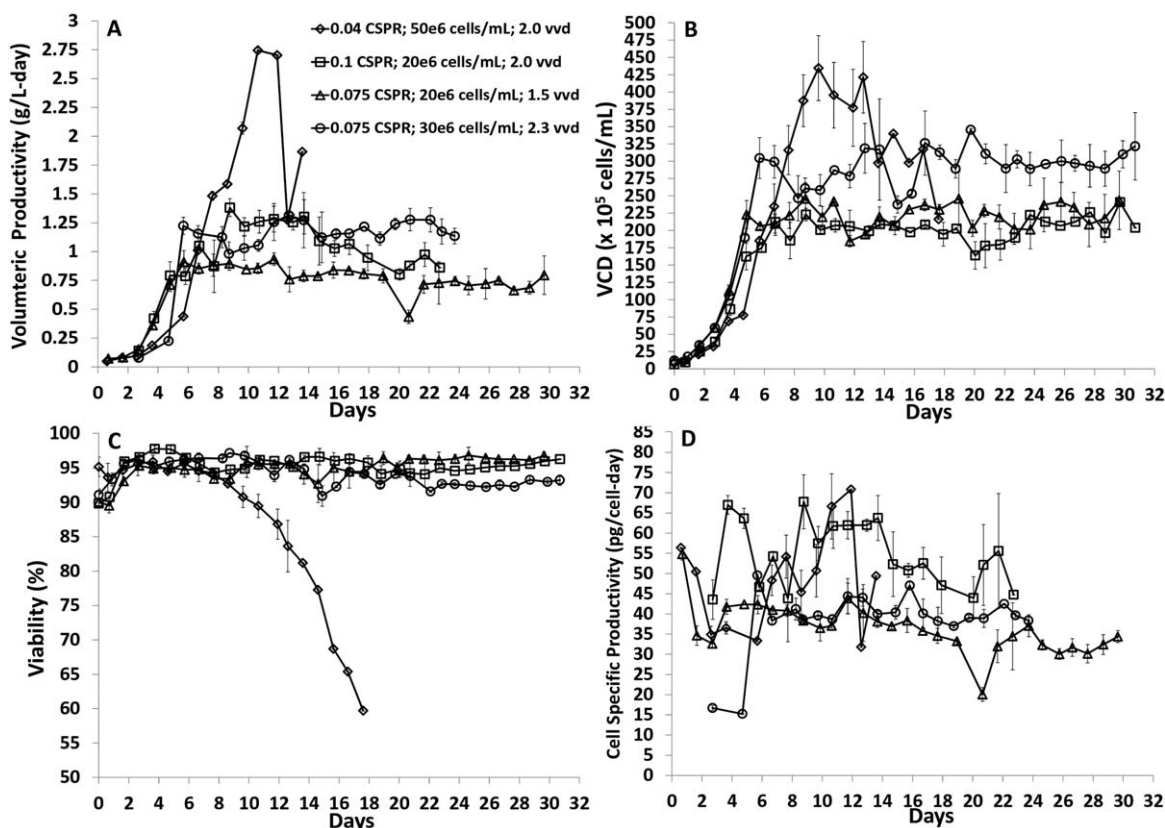


Figure 7. Establishing the critical CSPR for the high-yield perfusion medium. All conditions were performed in duplicates. The values indicated in the legend are targets; therefore, actual values vary with the control of VCD. Not all titer measurements are available for duplicate runs; therefore, some runs have incomplete productivity averages: (A) volumetric productivity (Q_p), (B) VCD, (C) viability, and (D) cell-specific productivity.

medium condition had duplicate runs and the 90:10 medium condition had a single run. The runs were not planned to go beyond the durations shown in the results as the intent was only to obtain proof of concept in the bioreactor settings. The perfusion culture VCD was controlled at 40×10^6 cells/mL. Viability profiles were comparable between both media (data not shown). The results clearly show that the performance improvements observed in deep-well plate models can be replicated in a large-scale bioreactor setting. Cell-specific productivity was improved after day 6 with the high-yield perfusion medium compared to the 90:10 medium (Figure 6A). The volumetric productivity (total produced including cell bleed) also reached higher levels after day 8 (Figure 6B). The reason the volumetric productivity ramped up more slowly in the high-yield perfusion medium is because the cell growth is designed to be slower than that of the 90:10 medium. The 90:10 medium reached 40×10^6 cells/mL on day 6, whereas the high-yield perfusion medium did not reach 40×10^6 cells/mL until after day 8 (data not shown). A future consideration is to use a growth promoting medium in the ramp-up phase and then switch over to the high-yield perfusion medium as the target VCD is reached. The observed lactate and ammonium reductions in the deep-well plate model were also replicated in the bioreactor (Figure 6C,D). The reductions were similar in magnitude in both types of culture vessels. Similar performance improvements in 2 L ATF bioreactors were also observed for the original cell line used to perform the medium development (data not shown). These 2 L bioreactor results confirm the improvements made in the high-yield perfusion medium.

Establishing the critical CSPR of the high-yield perfusion medium for sustainable perfusion

The same cell line of interest as described in the previous section was used to demonstrate sustainable pseudo steady-state perfusion culture with the high-yield perfusion medium. The goal was to obtain the lowest perfusion rate possible with this medium, that is, critical cell-specific perfusion rate (CSPR), while still having a sustainable high viability perfusion culture. Konstantinov et al. termed this perfusion medium characteristic as medium depth and can be quantified as the inverse of the minimum CSPR representing the maximum cell density that can be supported by a unit perfusion rate.²⁹ The concept of push-to-low perfusion rate was applied as described by Konstantinov et al.²⁹ At the start, it was determined that to plan for a viable facility fit of the perfusion process in our facility, the maximum perfusion rate allowed cannot exceed 2 medium volumes per bioreactor volume per day (vvd). The first replicate conditions targeted 0.04 nL/cell-day CSPR at 50×10^6 cells/mL and 2.0 vvd. In this condition, the maximum VCD achievable for this cell line was $\sim 46 \times 10^6$ cells/mL before the culture started declining in VCD and viability (Figures 7B,C). No cell bleed was used for this condition. The culture, however, showed it was capable of reaching a peak volumetric productivity (Q_p) of ~ 2.8 g/L-day (Figure 7A). The results concluded that the CSPR of ~ 0.04 nL/cell-day cannot sustain the perfusion culture.

Next was to determine the highest VCD that the 2 vvd perfusion rate could sustain. Cell densities of 30×10^6 cells/mL and 20×10^6 cells/mL were controlled via modulating

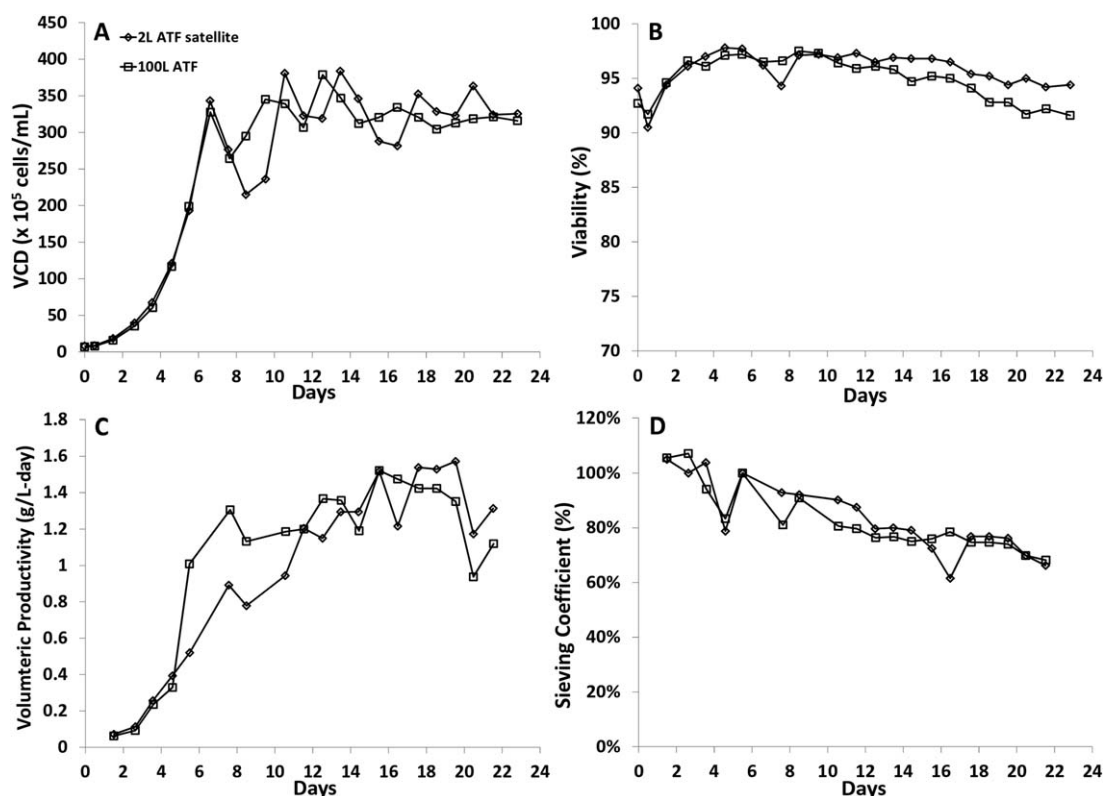


Figure 8. Scale-up comparison of 2 L ATF bioreactor to 100 L ATF bioreactor using the same inoculum, same lot of perfusion medium, and the same process set points: (A) VCD, (B) viability, (C) volumetric productivity, and (D) sieving coefficient, defined as the product concentration of the permeate stream divided by the product concentration inside the bioreactor.

cell bleed rate under a constant 2 vvd perfusion rate after ramp up. It was determined that the 30×10^6 cells/mL culture was not sustainable at the CSPP of 0.067 nL/cell-day (data now shown). However, the 20×10^6 cells/mL culture was sustainable with high viability for >30 days at the CSPP of 0.1 nL/cell-day (Figure 7C). At 20×10^6 cells/mL, however, the peak Q_p (accounting for total produced including cell bleed) decreased to ~ 1.2 g/L-day (Figure 7A).

To further determine whether the 0.1 nL/cell-day is really the critical CSPP, the perfusion culture was held at 20×10^6 cells/mL and the perfusion rate was tested at 2, 1.5, and 1.0 vvd equating to 0.1, 0.075, and 0.05 nL/cell-day CSPP, respectively. The results showed that the 0.05 nL/cell-day CSPP was not capable of sustaining a steady-state perfusion culture (data not shown) confirming the previous failed values of 0.04 and 0.067 nL/cell-day. However, the results showed that the 0.075 nL/cell-day CSPP was sufficient for sustaining a steady-state perfusion culture (Figure 7C) indicating that 0.01 nL/cell-day was not the critical CSPP needed for this cell line using the high-yield perfusion medium. Under 0.075 nL/cell-day at 20×10^6 cells/mL, only ~ 0.9 g/L-day peak Q_p was achieved (Figure 7A). This implies that at the same cell density, a lower CSPP even if capable of sustaining steady-state culture can still result in lower volumetric productivity than a higher sustaining CSPP. This is evident in Figure 7D showing a lower cell-specific productivity for 0.075 nL/cell-day than 0.01 nL/cell-day CSPP with both maintaining VCD of approximately 20×10^6 cells/mL. Therefore, there would be additional consideration of whether operating at higher CSPP than the critical level can be a better return on cost of goods given that higher productivity may be obtained.

After determining the critical CSPP is 0.075 nL/cell-day (very close to the failed CSPP of 0.067 nL/cell-day), the strategy is to increase the Q_p by pushing VCD higher while holding at this critical CSPP level. Perfusion rates of 1.5, 2.3, and 2.8 vvd paired respectively with 20, 30, and 40×10^6 cells/mL all equating to 0.075 nL/cell-day were compared. The perfusion culture was sustainable at all three perfusion rate to VCD paired conditions (data shown for only 2.3 vvd at 30×10^6 cells/mL) (Figure 7C). At 30×10^6 cells/mL, the peak Q_p improved to ~ 1.2 g/L-day compared to ~ 0.9 g/L-day at 20×10^6 cells/mL (Figure 7A). The cell-specific productivity at 0.075 nL/cell-day CSPP for 30×10^6 cells/mL remained similar to that for 20×10^6 cells/mL (Figure 7D). For this cell line, it was decided that 2.3 vvd can still be acceptable for demonstration at the large-scale 100 L pilot plant bioreactor. However, any perfusion rate higher than this level would be challenging to support in our existing pilot plant. Because we wanted to push for the highest Q_p possible for sustainable perfusion of this cell line, the 30×10^6 cells/mL at 2.3 vvd process was selected for demonstration of scale-up to 100L bioreactor.

Demonstrating the scalability and facility fit of perfusion process utilizing the high-yield perfusion medium

Large medium volume preparation in our existing fed-batch pilot plant is limited to the maximum of two 650 L batches per day to serve all pilot plant bioreactors. This is due to limitation of personnel, equipment capacity, and floor and storage space. Though medium volume required for perfusion process is inarguably much more than for fed-batch process, the recent cost of goods analysis demonstrated by Xu et al. shows that the medium cost in dollars per gram of

mAb produced can actually be similar between the two processes.³⁰ The reason is because the higher bioreactor productivity of the perfusion process offsets the higher medium cost it incurs. To conduct the perfusion process in 100 L single-use bioreactor using ATF6 at 2.3 vvd, >230 L of medium would be needed per day at steady state. Despite the fact that there is capacity to support this volume, this would not be practical as multiple projects share this capacity. For this reason, prepackaged sterile ready-to-use liquid medium direct from the vendor was developed. Conversion from powder to liquid medium product required additional development; details will be of a separate topic not covered in this article. Overall, considerations include the vendor making liquid from first principles rather than from medium powder, maintaining pH, and pCO₂ targets in sealed packaged environments rather than in large preparation vessels, manufacture at different scales, medium stability at transport temperatures, packaging design for plug-and-play use, and logistics planning to mitigate storage limitation and manage the short shelf-life of liquid products.

The pilot plant run was planned for 23 days. To support this, 5000 L of medium was made by the vendor in one single lot. The packaging was in 10 500 L bags held in portable totes with bottom openings allowing direct tubing connection to the 100 L SUB. The 10 totes were staged in an assembly line manner where each tote would be switched on to the bioreactor for feeding, then switched off to become the permeate collection vessel when empty. Two-liter ATF bioreactor was performed in parallel as a satellite run to demonstrate scale-up performance. The results showed that the scale-up of the ATF perfusion process using the high-yield perfusion medium performed similarly between the 2 L and 100 L scales. VCD, viability, and volumetric productivity profiles were comparable (Figures 8A–C). Notably, the sieving coefficient profiles were also similar and predictable despite the expected decline (Figure 8D). The performance at 100 L scale confirms that CSPR of 0.075 nL/cell-day can support a sustainable perfusion culture. The high-yield perfusion medium preparation was also scalable, and moreover, conversion to a liquid product yielded similar performance as a powder product (comparing Figure 8 results to Figure 7).

Conclusion

Perfusion medium can be created from mixtures of established fed-batch basal and feed media. Successful ratios can be obtained to serve as basis for further targeted optimization. Importance should be placed on understanding osmolality requirements as well as opportunity to remove redundant components to create more chemical space for further concentrating the perfusion medium. Ultimately reducing the number of components and concentrating the perfusion medium can support the use of lower perfusion rates and achieve lower critical CSPRs for sustainable perfusion cultures. One example is to ascertain which vitamins are important for the perfusion culture and to remove them if they have no benefits to performance. Subsequently, concentrations of beneficial vitamins and lipids can be increased to improve medium depth to support high cell density.

One development aspect of perfusion medium that differs from fed-batch medium is that sometimes, high cell growth is undesirable because the cell retention device already helps with mechanically achieving a high cell density culture. Uncontrolled cell growth can result in unproductive cell

mass and cause underperformance of q_p . In addition, the product loss from the cell bleed needed to maintain a certain high cell density can be substantially high (e.g., 10–30% of perfusion rate). This product loss is typically nonrecoverable. Therefore, it is of interest to be able to control cell growth and, hence, in turn potentially increase the q_p of the cells and reduce cell bleed. In principle, harvestable volumetric productivity improvement can be decoupled into three approaches: fundamentally improving the q_p , further increasing the cell density, and reducing the cell bleed use in order to minimize the product loss. Rebalancing Q, E, N, and D amino acids was discussed as an effective way of improving q_p by, in theory, shifting amino acid metabolism toward more secondary biosynthesis and less primary metabolism. To elicit such a response, concentrations in the medium were shifted to higher E and D, and lower Q and N levels.

Development of a perfusion medium also requires establishing the critical CSPR for specific cell lines using the medium. This is qualified as the lowest perfusion rate possible to sustain a high-viability steady-state culture. In some perfusion process formats where steady state is not required, this value is less important and the perfusion rate would be normally set based on the medium volume usage allowed and the target level of process intensification desired. In the example described in this article, the critical CSPR found for our cell line of interest using the high-yield perfusion medium was 0.075 nL/cell-day. However, it was also observed that going above the critical CSPR can further increase the cell-specific productivity of the cell line. In the end, the highest perfusion rate allowed based on the facility fit limit and the critical CSPR were used to determine the highest target steady-state cell density possible. The process was demonstrated to be successfully scalable from the 2L to the 100L bioreactor using ATF. Liquid medium product direct from the vendor was also developed based on the powder medium product to further mitigate facility capacity constraints, particularly the personnel resource and storage space. The liquid product was demonstrated to have equivalent performance to the medium made in house from powder product. Overall, this strategy saved significant human resources. The operation was also made more convenient by the use of plug-and-play medium totes. The cost including packaging and shipping was similar to manufacturing the medium in house.

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